

Artificial Intelligence & Machine Learning - Task 4

Breast Cancer Classification (Malignant vs Benign)

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Dataset & Methodology

Dataset Used:

- Breast Cancer Wisconsin Dataset (Built-in scikit-learn)
- Total Samples: 569
- Features: 30 numerical features
- Target: 0 = Malignant (Cancer), 1 = Benign
- Class Distribution: [Show the output of `np.bincount(y)`]

Steps Performed:

1. Data Loading & Exploration
2. Train-Test Split (with `stratify=y`)
3. Feature Scaling using `StandardScaler`
4. Trained Logistic Regression (Baseline + Balanced)
5. Trained Decision Tree Classifier
6. Evaluated using Confusion Matrix, Precision, Recall, F1-Score, ROC-AUC

Results & Visualizations

Key Results:

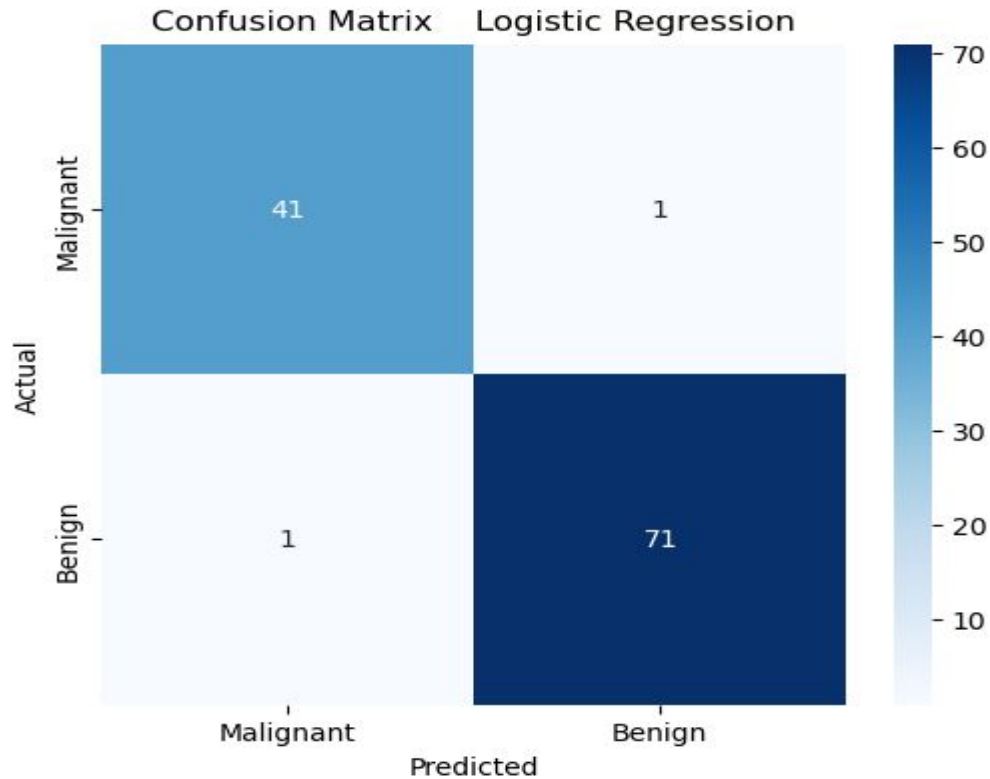
Accuracy: 0.9824561403508771

Classification Report:

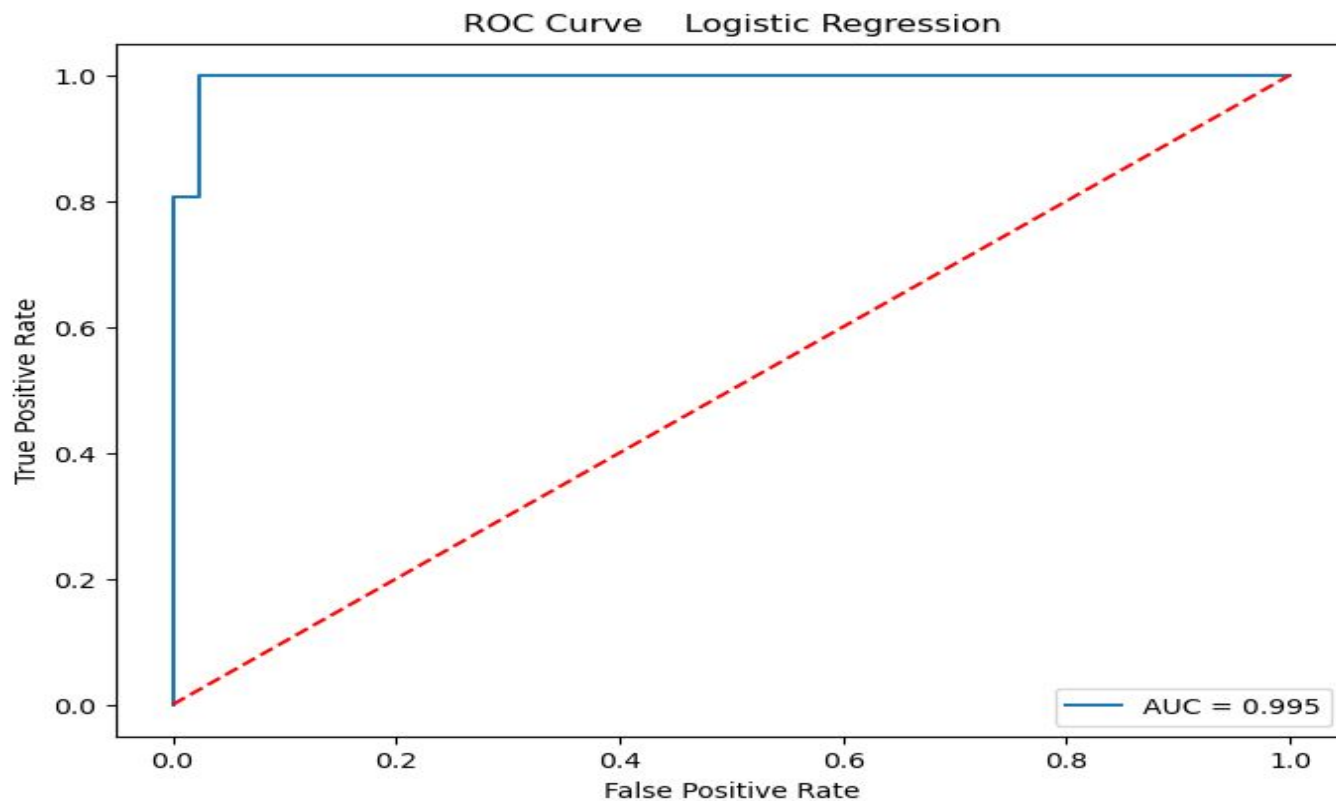
	precision	recall	f1-score	support
0	0.98	0.98	0.98	42
1	0.99	0.99	0.99	72
accuracy			0.98	114
macro avg	0.98	0.98	0.98	114
weighted avg	0.98	0.98	0.98	114

	Model	Accuracy	AUC
0	Logistic Regression	0.982456	0.995370
1	Balanced LR	0.956140	0.995370
2	Decision Tree	0.912281	0.915675

Results & Visualizations



Results & Visualizations



Analysis, Justification & Conclusion

Metric Selection Justification:

- Accuracy alone is not sufficient because the dataset has class imbalance.
- **Recall** is critical in medical diagnosis — We cannot afford to miss Malignant cases (False Negative is dangerous).
- **F1-Score** gives good balance between Precision and Recall.
- **ROC-AUC** is used to measure overall model discrimination ability.

Handling Imbalanced Data:

- Used `class_weight='balanced'` in Logistic Regression.
- This improved Recall for the minority class (Malignant).

Final Model Selection:

- **Best Model:** Logistic Regression (with `class_weight='balanced'`)
- Reason: Highest AUC score, better Recall, stable performance, and good generalization compared to Decision Tree.